# Numerical Simulations of the SVIR Model with Vaccination: Bistability and the Basin Separatrix (prepared prompting Claude AI)

**Abstract**

We integrate numerically an SVIR model with waning vaccine protection and waning immunity, with a vaccination rate $\phi$ chosen so that the controlled reproduction number $r$ lies in the bistability window $(r_2, 1)$ predicted by the quadratic equilibrium analysis. For $\phi = 0.0085$ ($r \approx 0.9813$) the disease-free equilibrium and an endemic equilibrium ($I \approx 0.0237$) are both stable. Initial conditions with no vaccinated individuals converge to the endemic state, initial conditions near the vaccinated disease-free state may converge to either, and we compute the curve in the $(V_0, I_0)$ plane that separates the two basins of attraction.

## 1 Model and parameters

### 1.1 Equations

The population is normalized to $N = 1$ and divided into susceptible $S$, vaccinated $V$, infected $I$ and recovered $R$ individuals:

$$\begin{aligned} \dot{S} &= \mu N - \beta S I - \mu S - \phi S + \theta V + \psi R, \\ \dot{V} &= \phi S - (\mu + \theta) V - \sigma \beta V I, \\ \dot{I} &= \beta S I + \sigma \beta V I - \gamma I - \mu I, \\ \dot{R} &= \gamma I - (\mu + \psi) R. \end{aligned} \tag{1}$$

Adding the four equations gives $\dot{S} + \dot{V} + \dot{I} + \dot{R} = \mu\big(N - S - V - I - R\big)$, so $S + V + I + R = 1$ is invariant. Here $\phi$ is the vaccination rate, $\theta$ the rate of loss of vaccine protection, $\sigma = 1 - \epsilon$ the relative susceptibility of vaccinated individuals ($\epsilon$ is the vaccine efficacy), and $\psi$ the rate of loss of natural immunity.

### 1.2 Parameter values

| Parameter | Value |
|---|---|
| $\mu$ | $360 \times 10^{-7}$ |
| $\gamma$ | $1/7$ |
| $\psi$ | $1/90$ |
| $\theta$ | $1/1095$ |
| $\epsilon$, $\sigma$ | 0.8, 0.2 |
| $\mathcal{R}_0$ | 3.5 |
| $\beta$ | $\mathcal{R}_0(\mu + \gamma) \approx 0.500126$ |
| $\phi$ | 0.0085 (main runs); 0.00854 (preliminary runs) |

### 1.3 Disease-free equilibrium and the controlled reproduction number

The disease-free equilibrium (DFE) is

$$S^* = \frac{\mu+\theta}{\mu+\theta+\phi}, \qquad V^* = \frac{\phi}{\mu+\theta+\phi}, \qquad I^* = R^* = 0. \tag{2}$$

Linearizing the $I$-equation at (2) gives $\dot{I} = (\gamma+\mu)(r-1)I$ with

$$r = \frac{\beta(S^*+\sigma V^*)}{\gamma+\mu} = \mathcal{R}_0 \frac{(\mu+\theta)+\sigma\phi}{\mu+\theta+\phi}, \qquad \text{equivalently} \qquad \phi = (\mu+\theta)\frac{\mathcal{R}_0 - r}{r - \mathcal{R}_0\sigma}. \tag{3}$$

The second relation is the function $\phi(r)$ used in the equilibrium analysis. The DFE is locally stable for $r < 1$. The endemic equilibria satisfy the quadratic $AI^2 + B(r)I + C(r) = 0$ studied in the companion analysis, where $\mathcal{R}_0\sigma = 0.7 < 1 < \mathcal{R}_0$.

For $\beta = 0.5$ the discriminant numerator $N$ has roots $r_1 \approx 0.728$ and $r_2 \approx 0.944$, and the reverse bifurcation is folded at $r_2$: for $r \in (r_2, 1)$ the DFE ($r < 1$) coexists with two endemic equilibria $I_1(r) > I_2(r) > 0$, the upper one stable. Both choices of $\phi$ used below correspond to $r$ in this window:

- $\phi = 0.0085$ gives $r \approx 0.9813$ by (3);
- the preliminary runs used $r = 0.98$, i.e. $\phi = 0.008543$ (this value was used before the value 0.0085 was specified).

At $\phi = 0.0085$, $S^* \approx 0.1005$, $V^* \approx 0.8995$, and the endemic equilibrium has $I \approx 0.02367$.

### 1.4 Numerical method

System (1) was integrated on $t \in [0, 3000]$ with the LSODA method (scipy `solve_ivp`, tolerances $10^{-10}$ relative and $10^{-13}$ absolute). In all runs the conserved total $S + V + I + R$ stayed equal to 1 to machine accuracy. Curves are colored: $S$ green, $V$ orange, $I$ red, $R$ blue.

## 2 Runs with no vaccinated individuals at $t = 0$

The two initial conditions are

$$\text{(i) } S_0 = 0.99,\ I_0 = 0.01, \qquad \text{(ii) } S_0 = 0.9,\ I_0 = 0.1, \qquad V_0 = R_0 = 0.$$

### 2.1 Preliminary runs ($r = 0.98$, $\phi \approx 0.00854$)

As a first choice, $\phi$ was computed from $r = 0.98$, which lies inside the bistability window $(r_2, 1)$.

*Comments.* Both runs converge to the same endemic state, $S \approx 0.188$, $V \approx 0.488$, $I \approx 0.02346$, $R \approx 0.301$. This value of $I$ agrees with the larger root $I_1(0.98) = 0.023456$ of the equilibrium quadratic, which checks the consistency of the simulation with the algebraic analysis. The infection peak is $I \approx 0.324$ at $t = 16$ in run (i) and $I \approx 0.374$ at $t = 8.5$ in run (ii), followed by damped oscillations. Even much smaller initial infections ($I_0$ down to $10^{-6}$, still with $V_0 = 0$) end at the endemic state.

### 2.2 Main runs ($\phi = 0.0085$, $r \approx 0.9813$)

*Comments.* Again both runs end at the same endemic equilibrium: at $t = 3000$, $S \approx 0.1889$, $V \approx 0.4840$, $I \approx 0.02367$, $R \approx 0.3034$, to be compared with the equilibrium value $I = 0.023670$ from the quadratic. The infection peaks at $I \approx 0.324$ ($t = 16$) in run (i) and $I \approx 0.374$ ($t = 8.5$) in run (ii). Starting from $V_0 = 0$ the system is far from the vaccinated DFE, so it always reaches the endemic state in these experiments.

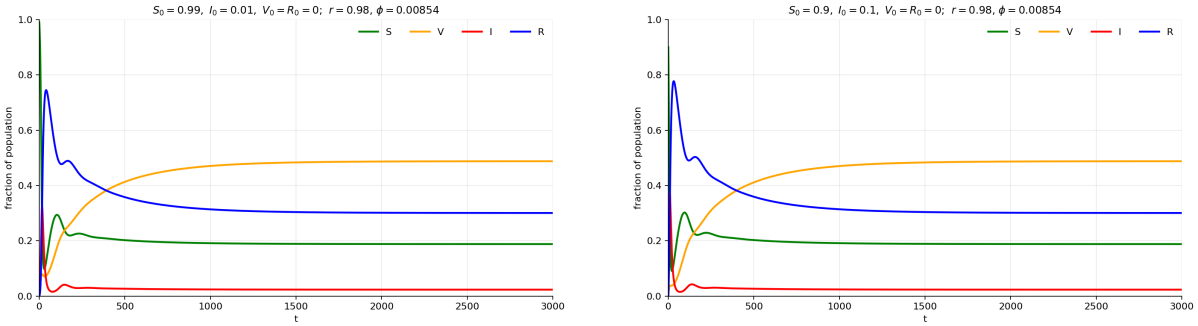



Figure 1: Preliminary runs, $r = 0.98$ ($\phi \approx 0.00854$). Left: initial condition (i). Right: initial condition (ii).

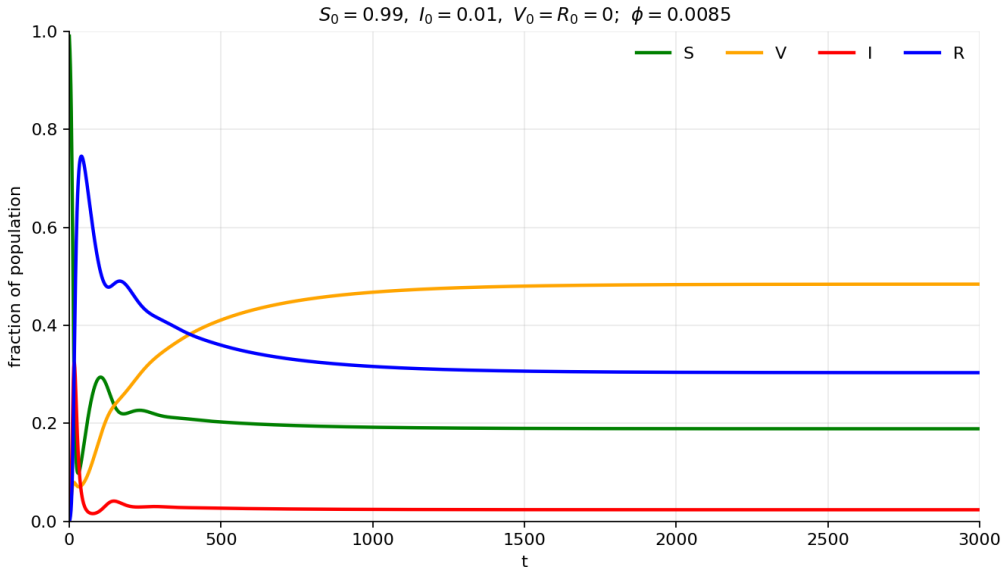



Figure 2: $\phi = 0.0085$, initial condition (i): $S_0 = 0.99$, $I_0 = 0.01$, $V_0 = R_0 = 0$.

## 3 The disease-free alternative

To see the other attractor we start at the vaccinated DFE (2) with a small infection: $S_0 = S^*(1 - I_0) \approx 0.1004$, $V_0 = V^*(1 - I_0) \approx 0.8986$, $I_0 = 0.001$, $R_0 = 0$ (so that $S_0 + V_0 + I_0 = 1$).

*Comments.* $S$ and $V$ remain near $S^* \approx 0.1005$ and $V^* \approx 0.8995$, while $I$ decays exponentially from $10^{-3}$. The slope in the log plot matches the linearization at the DFE, $(r - 1)(\gamma + \mu) \approx -0.002675$ per unit time. At $t = 3000$, $I \approx 7.5 \times 10^{-7}$ and $R \approx 1.3 \times 10^{-5}$; $R$ first rises to about $8 \times 10^{-3}$ and then decays as well. Starting from the same state with a larger infection, $I_0 = 0.01$, the system instead converges to the endemic state ($I \approx 0.0237$), so the two attractors coexist, as predicted by the reverse bifurcation.

## 4 A run starting from a highly vaccinated population

Take $V_0 = 0.9$, $I_0 = 0.01$, $R_0 = 0$, $S_0 = 1 - (V_0 + I_0) = 0.09$.

*Comments.* Although the initial effective reproduction number $\beta(S_0 + \sigma V_0)/(\gamma + \mu) \approx 0.945$ is below 1, the run does *not* end at the DFE. The infection first decreases, to a minimum $I \approx 0.0030$ near $t \approx 182$, and then increases slowly toward the endemic equilibrium: $I = 0.02353$ at $t = 3000$ and $I = 0.023670$ at $t = 30000$ (with $V \approx 0.484$, $S \approx 0.189$, $R \approx 0.303$ in the long run). The

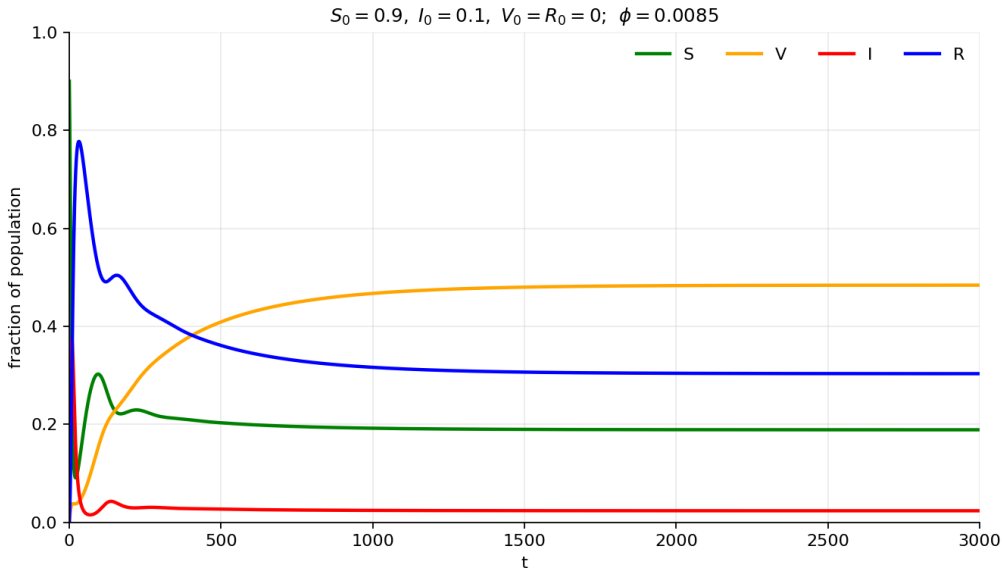



Figure 3: $\phi = 0.0085$, initial condition (ii): $S_0 = 0.9$, $I_0 = 0.1$, $V_0 = R_0 = 0$.

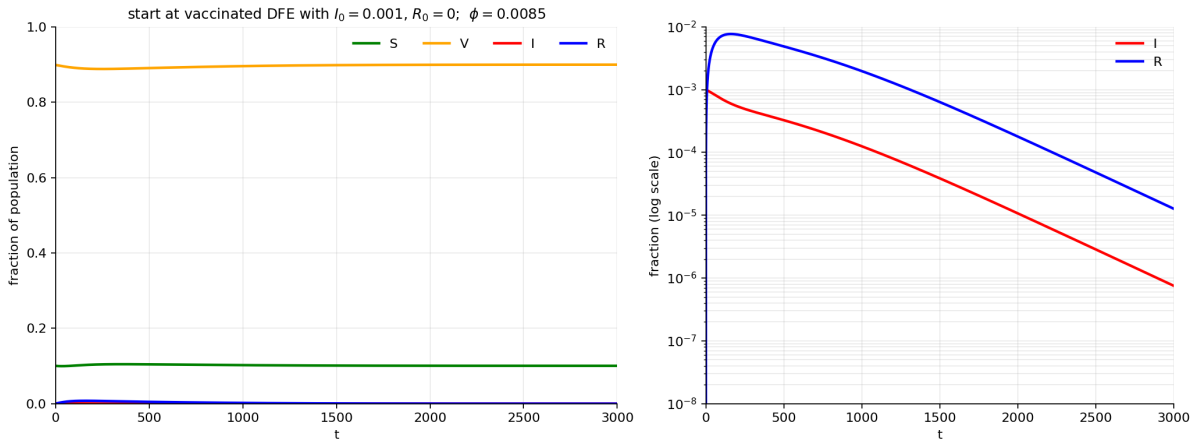



Figure 4: Disease-free alternative, $\phi = 0.0085$. Left: all compartments (the curves $I$ and $R$ are too small to see at this scale). Right: $I$ and $R$ on a logarithmic scale.

initial dip is a transient: recovered individuals return to the susceptible class through waning immunity (rate $\psi$), the vaccinated class is depleted through waning vaccine protection (rate $\theta$) and through infection, and the larger susceptible pool allows the infection to grow again. This point therefore lies on the endemic side of the basin boundary computed next.

# 5 Basin of attraction of the two equilibria

## 5.1 Slice and method

The state space is three-dimensional ($S + V + I + R = 1$), so a picture in the $(V_0, I_0)$ plane requires fixing a slice. We use the slice of the previous runs:

$$R_0 = 0, \qquad S_0 = 1 - V_0 - I_0, \qquad \phi = 0.0085.$$

For each $V_0 \in [0.65, 0.99]$ (step 0.0025) the smallest $I_0$ that leads to the endemic equilibrium was found by bisection on full trajectories (relative tolerance $2 \times 10^{-4}$). A trajectory was classified

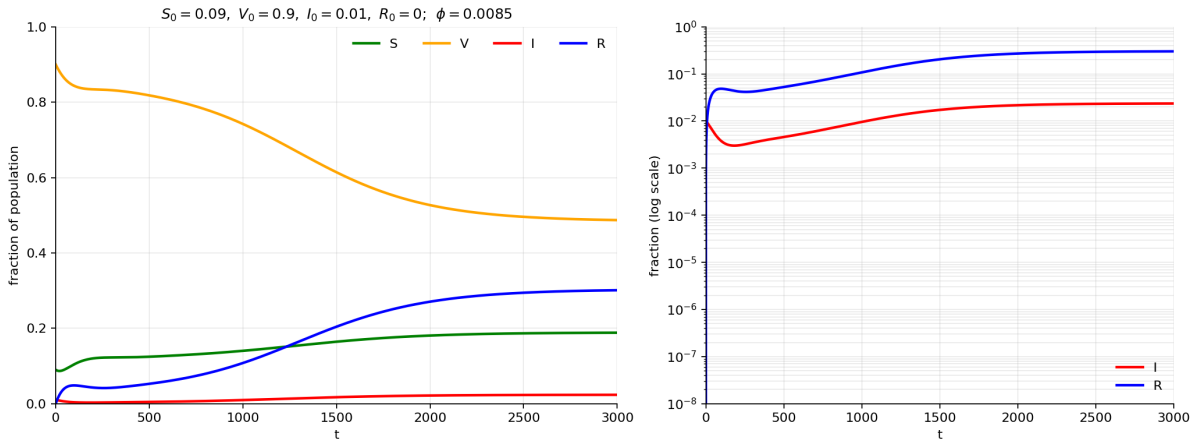



Figure 5: $\phi = 0.0085$, initial condition $S_0 = 0.09$, $V_0 = 0.9$, $I_0 = 0.01$, $R_0 = 0$. Left: all compartments. Right: $I$ and $R$ on a logarithmic scale.

as disease-free if $I$ dropped below $10^{-9}$, and as endemic if $I > 10^{-4}$ at $t = 80000$ (maximum integration time). For $V_0 = 0.86, 0.90, 0.94, 0.96$ a scan of 200 values of $I_0$ showed exactly one switch from disease-free to endemic, so the boundary is a single threshold $I_c(V_0)$ in $I_0$ there. A coarse grid scan over $V_0 \in [0.65, 0.99]$, $I_0 \in [0, 0.03]$ showed the same pattern.

### 5.2 Result

Selected values of the threshold are

| $V_0$ | $I_c(V_0)$ |
|---|---|
| 0.65 | $2.0 \times 10^{-7}$ |
| 0.80 | $9.0 \times 10^{-5}$ |
| 0.90 | $3.8 \times 10^{-3}$ |
| 0.95 | $1.83 \times 10^{-2}$ |

*Comments.*

- The threshold increases steeply with $V_0$; on a logarithmic scale it is nearly a straight line, so $I_c$ grows roughly exponentially with $V_0$. It is strictly positive for all $V_0$ shown (about $2 \times 10^{-7}$ at $V_0 = 0.65$), so on the linear scale of Figure 6 it hugs the horizontal axis for $V_0 \lesssim 0.85$.
- The more vaccinated the initial population, the larger the infection needed to reach the endemic state. Outside the range plotted, the curve reaches $I_0 = 0.03$ at $V_0 \approx 0.968$.
- The earlier test points are consistent with the curve: $I_c(0.9) \approx 0.0038$, so $I_0 = 0.001$ falls in the disease-free basin and $I_0 = 0.01$ in the endemic one.
- The curve depends on the slice $R_0 = 0$ and on the value of $\phi$; other choices would shift it.

## 6 Summary

1. With $\phi = 0.0085$ ($r \approx 0.9813 \in (r_2, 1)$) the model has two stable equilibria: the vaccinated disease-free state ($S^* \approx 0.1005$, $V^* \approx 0.8995$) and an endemic state ($S \approx 0.189$, $V \approx 0.484$, $I \approx 0.0237$, $R \approx 0.303$). The endemic value of $I$ agrees with the root $I_1(r)$ of the equilibrium quadratic.

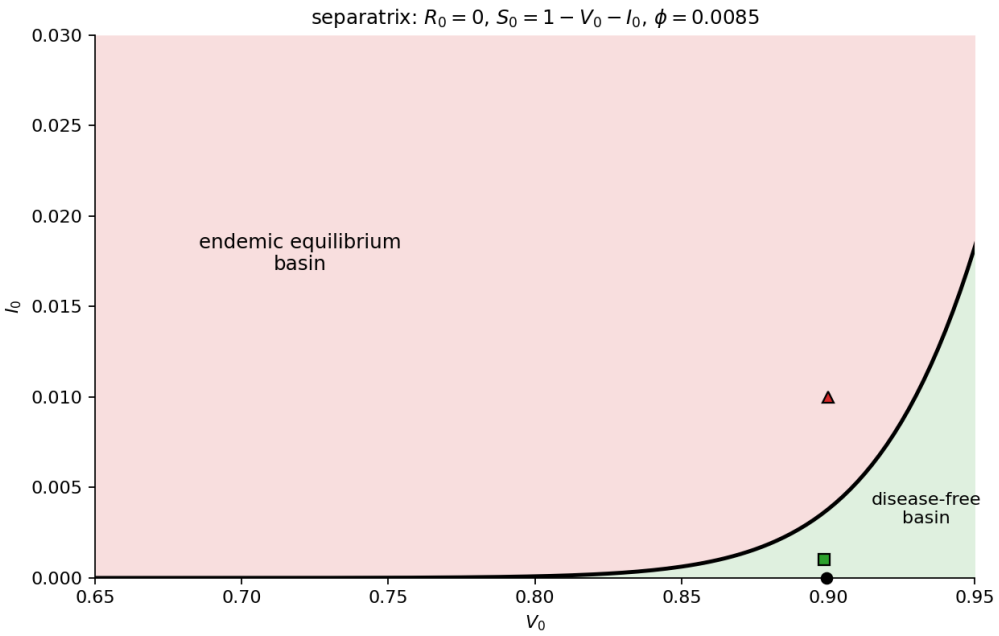



Figure 6: Separatrix between the basins of attraction in the $(V_0, I_0)$ plane, for $R_0 = 0$, $S_0 = 1 - V_0 - I_0$, $\phi = 0.0085$. Above the black curve (red) trajectories go to the endemic equilibrium; below it (green) to the disease-free equilibrium. The black dot is the vaccinated DFE ($V^* \approx 0.8995$). The green square $(0.8986, 0.001)$ and the red triangle $(0.9, 0.01)$ are initial conditions from Sections 3 and 4 that went to the DFE and to the endemic state, respectively.

2. Initial conditions with $V_0 = 0$ always converged to the endemic state in the experiments, even for very small $I_0$.

3. Starting at the vaccinated DFE with $I_0 = 0.001$ the infection dies out; with $I_0 = 0.01$ it reaches the endemic state.

4. From $V_0 = 0.9$, $I_0 = 0.01$ the infection first declines and then rebounds to the endemic state.

5. The basin separatrix in the $(V_0, I_0)$ plane (slice $R_0 = 0$) is a single increasing curve $I_c(V_0)$, from about $2 \times 10^{-7}$ at $V_0 = 0.65$ to $1.83 \times 10^{-2}$ at $V_0 = 0.95$.